



Machine Learning Concepts



Course Objectives

Introduction to Machine Learning

Software Development vs Machine Learning

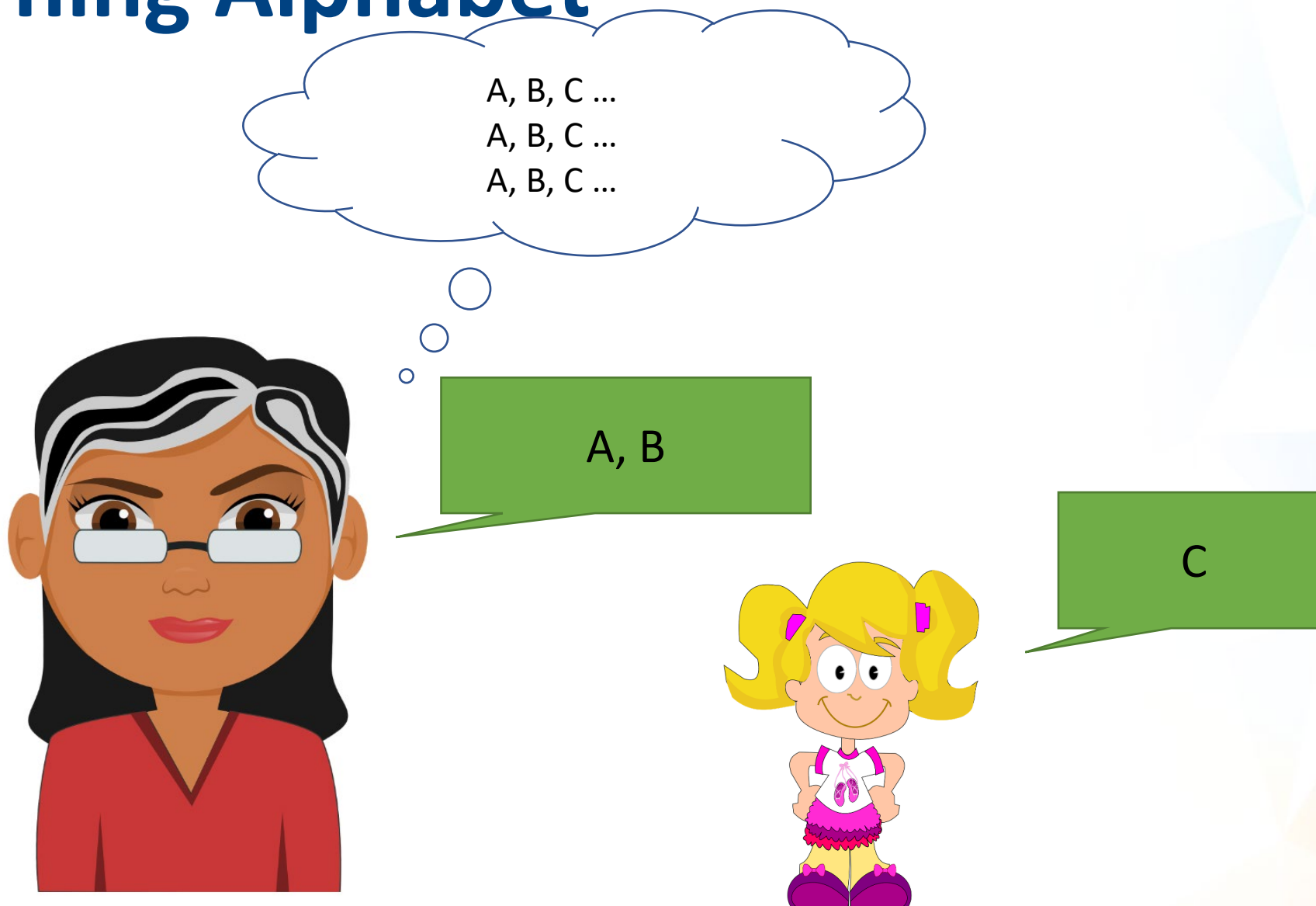
Broad Categories of Machine Learning

Learning Outcomes

At the end of the course, you will be able to

- Understand the concepts of machine learning
- Differentiate the difference between software development and machine learning
- Aware of the two broad categories of machine learning

Learning Alphabet



How does a child learn?

Identifying the needs

Get a book

Reciting the alphabets

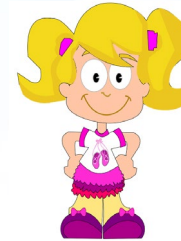
Pattern Discovery / Learning takes place

Prediction



A, B, C ...
A, B, C ...
A, B

C



What is Machine Learning?

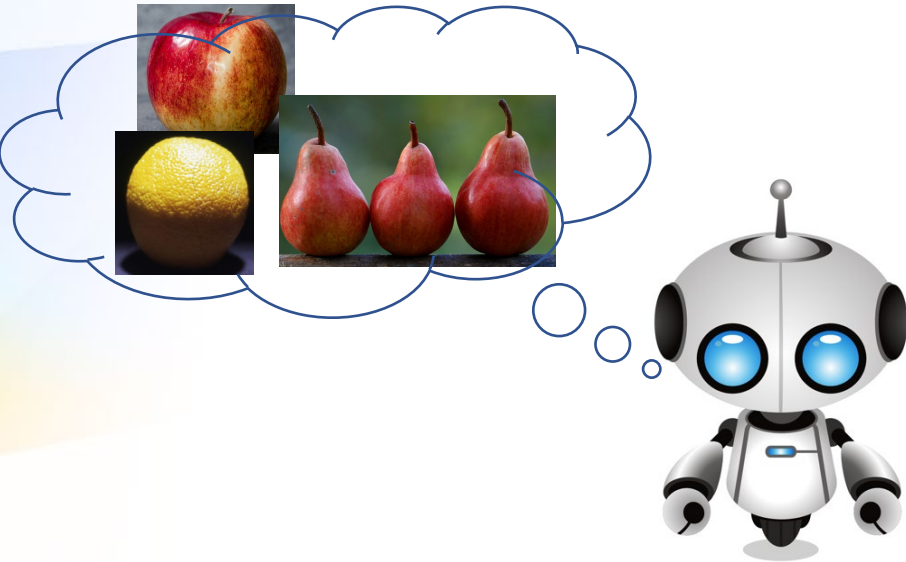
Problem
Statement

Data
Preparation

Feeding
of data

Pattern Discovery /
Learning takes place

Prediction



Software Development vs Machine Learning

Software Development

Developers
code the rules
into the
software



Example: Coded Business Rules

- Minimum fixed charge: \$2.40
- Fare is computed as \$0.40 per kilometer
- X = Distance travelled

$$\text{Taxi Fare } (y) = \$2.40 + x * 0.4$$

Provide x value, it will
return y value

Machine Learning

Machine learns
from the
dataset



Example: Generated Prediction equation

$$Y = mX + b$$

(values of m and b are defined)

To perform predict Y based
on X , we provide value to X

Software Development vs Machine Learning

Software Development

Changes in biz
rules!



Developers
change the
code

Software



Updated
Features

Machine Learning

New Dataset



Machine
Learning Model



Updated Predication
Equation!

Machine Learning

Broad Categories

Both approaches require human intervention

- Supervised: to label data accurately
- Unsupervised: to validate output variables

Goals:

- Supervised: Predict outcomes for new data (you know what to expect)
- Unsupervised: Get insights from large volumes of new data (less predictable)

- Supervised Learning

- Classification

- Algorithms – Logistic Regression, KNN, Decision Tree, etc.

- Linear Regression

- Algorithms – Linear Regression

- Unsupervised Learning

- Clustering

- Algorithms – K-means, Hierarchical Clustering, DBScan, etc.

In supervised learning, the algorithm “learns” from the training data set by iteratively making predictions on the data and adjusting for the correct answer.

Unsupervised learning models, in contrast, work on their own to discover the inherent structure of unlabeled data.

Supervised Learning

Task – Sales Forecast

	A	B
1	Advert	Sales
2	1000	10000
3	2000	19000
4	3000	29000
5	4000	46000
6	5000	53000
7	6000	67000
8	7000	71110

X

(feature)

Y

(label / target outcome)

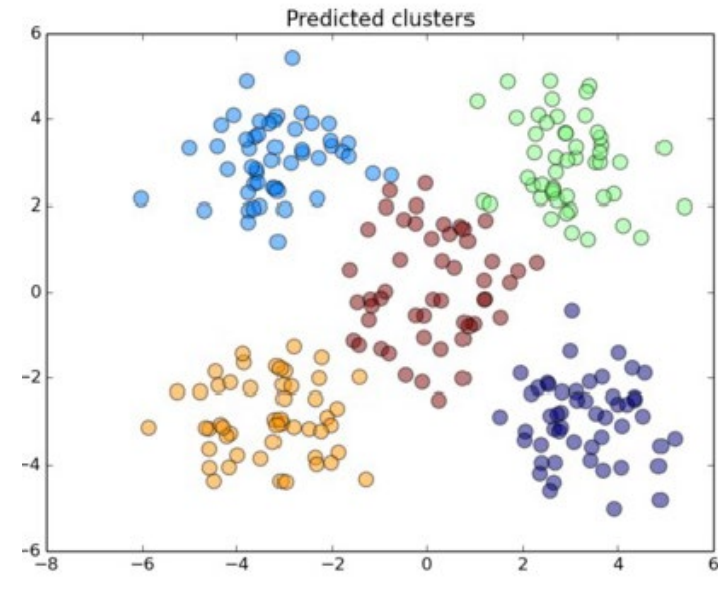
Send the data

Linear
Regression
Model

Capabilities:
Predict y
based on x

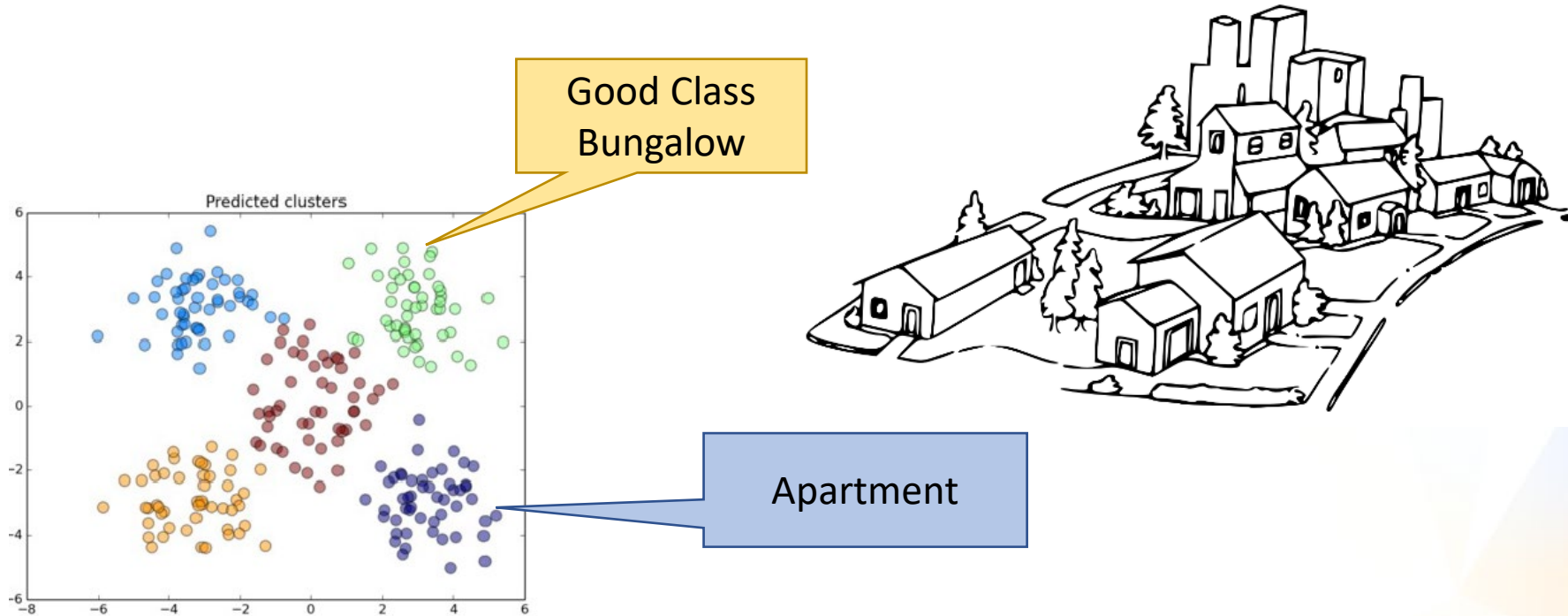
Unsupervised Learning

- Unlabelled data
- Allows the model to work on its own to discover patterns



Unsupervised Learning - Clustering

- finding natural groupings in a set of data
 - without pre specifying a set of categories.
 - Algorithms – K-means, Hierarchical Clustering, DBScan, etc.



What Have We Learnt?

- The concepts of machine learning
- The difference between software development and machine learning
- The two broad categories of machine learning